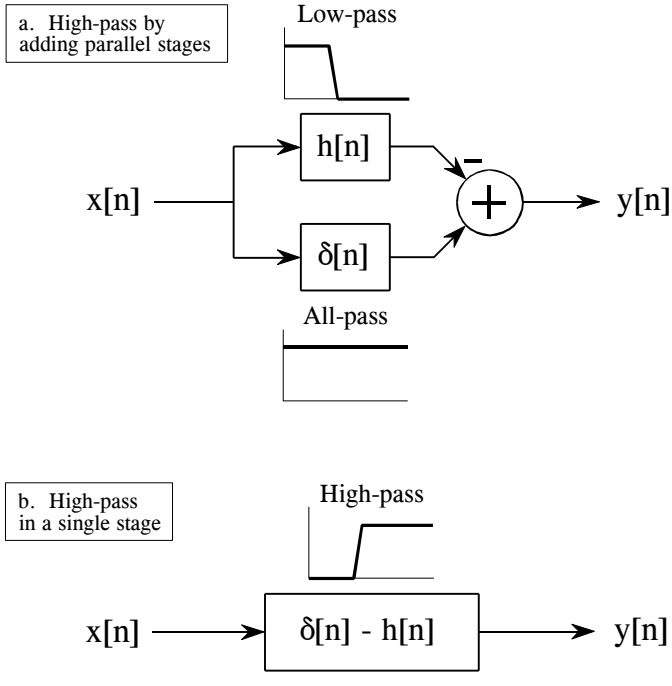

FIGURE 14-6
Block diagram of spectral inversion. In (a), the input signal, $x[n]$, is applied to two systems in parallel, having impulse responses of $h[n]$ and $\delta[n]$. As shown in (b), the combined system has an impulse response of $\delta[n] - h[n]$. This means that the frequency response of the combined system is the *inversion* of the frequency response of $h[n]$.

frequency response is shown in (b), found by adding 13 zeros to the filter kernel and taking a 64 point FFT. Two things must be done to change the low-pass filter kernel into a high-pass filter kernel. First, change the sign of each sample in the filter kernel. Second, add *one* to the sample at the center of symmetry. This results in the high-pass filter kernel shown in (c), with the frequency response shown in (d). Spectral inversion *flips* the frequency response *top-for-bottom*, changing the passbands into stopbands, and the stopbands into passbands. In other words, it changes a filter from low-pass to high-pass, high-pass to low-pass, band-pass to band-reject, or band-reject to band-pass.

Figure 14-6 shows why this two step modification to the time domain results in an inverted frequency spectrum. In (a), the input signal, $x[n]$, is applied to two systems in parallel. One of these systems is a low-pass filter, with an impulse response given by $h[n]$. The other system does *nothing* to the signal, and therefore has an impulse response that is a delta function, $\delta[n]$. The overall output, $y[n]$, is equal to the output of the all-pass system *minus* the output of the low-pass system. Since the low frequency components are subtracted from the original signal, only the high frequency components appear in the output. Thus, a high-pass filter is formed.

This could be performed as a two step operation in a computer program: run the signal through a low-pass filter, and then subtract the filtered signal from the original. However, the entire operation can be performed in a signal stage by combining the two filter kernels. As described in Chapter